



**SUNFLASER**  
OPTOELECTRONIC SOLUTIONS

# SFA2000A

Laser Rangefinder Module



## Product Manual

Version 202506V1

## Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

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## Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

## Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

## Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- › Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- › Continuing to operate the ranging module after improper handling or misuse;
- › Using a ranging module with obvious faults, defects, or damage;
- › Performing test operations within the blind zone of the ranging module.

### Improper use may result in the following risks:

- › Eye injury
- › Incorrect results
- › Measurement errors
- › Property damage
- › Equipment malfunction

### Operator Responsibilities

- › Be familiar with the basic knowledge required for the correct operation of the ranging module;
- › Understand and comply with the operation and maintenance requirements of the ranging module;
- › Prevent damage to the ranging module during storage and transportation;
- › Keep the ranging module clean and well maintained;
- › Ensure that the ranging module is stored and operated in a safe environment;
- › Do not expose the ranging module to strong mechanical shock or impact;
- › Do not expose the ranging module to environments with drastic temperature fluctuations.

### Avoid Measurement Errors

- › Pay attention to all factors that may affect the performance of the ranging module;

- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

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# 01

## Product Overview

The SFA2000A Laser Rangefinder Module is an in-house developed laser ranging product designed based on the TOF (Time of Flight) principle.

The module consists of a laser transmitting unit, transmitting optical assembly, receiving optical assembly, and main control circuit. It communicates with the host system via a UART (TTL\_3.3V) serial interface, while providing a complete serial communication protocol and command set for rapid integration and secondary development.

Featuring a compact size, lightweight structure, stable performance, high-impact resistance, and Class 1 eye-safe laser design, the product is widely suitable for applications such as civilian UAVs, outdoor observation, engineering surveying, and other optoelectronic systems.

# 02

## Key Features

- Basic Ranging

Supports both single-shot and continuous ranging modes, with the ability to stop ranging at any time.

➤ Device Self-Test

Monitors system status, bias voltage, and temperature during power-on and operation to ensure normal performance.

➤ Parameter Configuration

Supports baud rate configuration to accommodate different serial communication requirements.

➤ Debugging Function

Supports spot brightness adjustment (for development purposes only) and allows quick exit to restore normal operating mode.

➤ Version Information Query

Supports querying FPGA version, MCU version, and hardware version information.

➤ Device Information Management

Supports reading the device SN serial number for convenient management and traceability.

➤ Low-Power Control

Supports power on/off control through the POWER\_ON pin for low-power standby management.



# 03

## Key Specifications

| Item                       |                                                             | Parameter                                         |
|----------------------------|-------------------------------------------------------------|---------------------------------------------------|
| Performance Specifications | Eye Safety Class                                            | Class 1 (IEC 60825-1)                             |
|                            | Laser Wavelength                                            | 905 ±10 nm                                        |
|                            | Laser Beam Divergence (Typical)                             | 2.5 × 5 mrad                                      |
|                            | Receiver Field of View                                      | ~8 mrad                                           |
|                            | Maximum Measuring Range<br>(Visibility <sup>®</sup> ≥ 6 km) | ≥2000 m @ 60% reflectivity, building target       |
|                            |                                                             | ≥1400 m @ 30% reflectivity, 2.3 × 2.3 m target    |
|                            |                                                             | ≥1000 m @ 30% reflectivity, 0.5 × 1.7 m target    |
|                            |                                                             | ≥500 m @ 30% reflectivity, 0.2 × 0.3 m target     |
|                            | Minimum Measuring Range                                     | 2 m                                               |
|                            | Ranging Accuracy (Building Target)                          | d ≤ 800 m: ±(0.4 m + d × 0.07%)                   |
|                            |                                                             | d > 800 m: ±1 m (d represents distance in meters) |
|                            | Measurement Frequency                                       | 1–5 Hz (Adaptive)                                 |
| Electrical Specifications  | Detection Probability                                       | ≥98%                                              |
|                            | False Alarm Rate                                            | ≤1%                                               |
|                            | Interface Type                                              | UART (TTL_3.3V)                                   |
|                            | Supply Voltage                                              | DC 3–5 V                                          |
|                            | Standby Power Consumption                                   | ≤0.5 mW (POWER_ON pulled low)                     |
|                            |                                                             | ≤0.3 W (POWER_ON pulled high)                     |
|                            | Operating Power Consumption<br>(+5 V Input)                 | ≤0.70 W @ 25°C                                    |
|                            |                                                             | ≤0.75 W @ 70°C                                    |
|                            |                                                             | ≤0.60 W @ -20°C                                   |
|                            | Startup Time                                                | ≤200 ms                                           |

|                              |                       |                                |
|------------------------------|-----------------------|--------------------------------|
| Physical Characteristics     | Weight                | 10.5 ±0.5 g                    |
|                              | Dimensions            | 26 × 25 × 14 mm (L × W × H)    |
|                              | Shock Resistance      | 1200 g, 1 ms                   |
|                              | Vibration Resistance  | 5–50–5 Hz, 1 octave/min, 2.5 g |
| Environmental Specifications | Operating Temperature | -20°C to +70°C                 |
|                              | Storage Temperature   | -25°C to +75°C                 |
|                              | Reliability           | MTBF ≥1500 h                   |

**Note:**

① Visibility refers to atmospheric visibility under haze or fog conditions. In sandstorm environments, the ranging performance may be affected more significantly.

# 04

## Interface and Connection

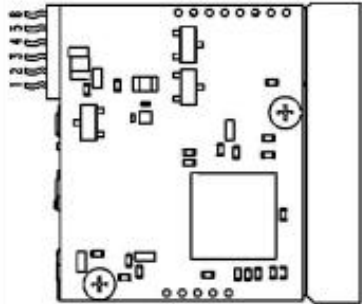
Communication Interface: UART (TTL 3.3 V), default baud rate: 115200 bps.

### Electrical Interface:

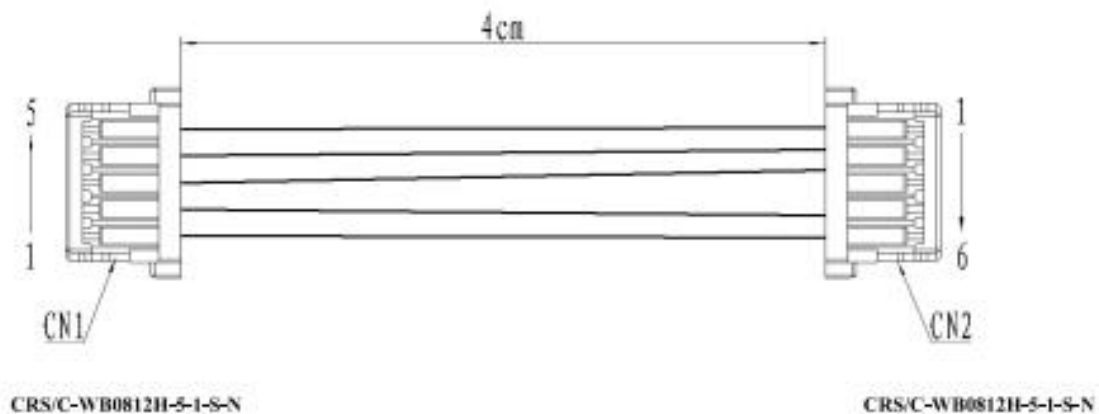
Connector model: FWF08002-S06B13W5M.

Refer to the table below for the connector pin definitions.

|       |                |                                                                            |
|-------|----------------|----------------------------------------------------------------------------|
| Pin 1 | POWER_ON       | Module power switch, TTL_3.3V level; Module on (>2.7V), module off (<0.3V) |
| Pin 2 | UART_RX        | Serial port receiver, TTL_3.3V level                                       |
| Pin 3 | UART_TX        | Serial port transmitter, TTL_3.3V                                          |
| Pin 4 | NC             |                                                                            |
| Pin 5 | Mains positive | Power supply, 3-5V                                                         |
| Pin 6 | GND            | Serial port grounding                                                      |



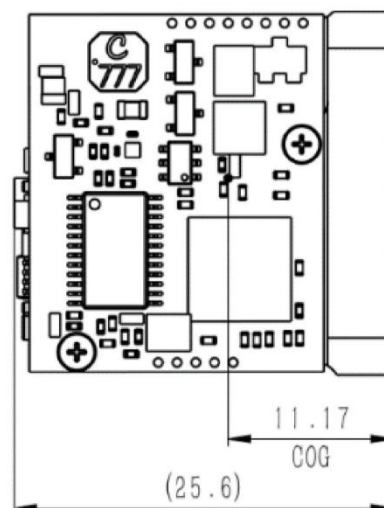
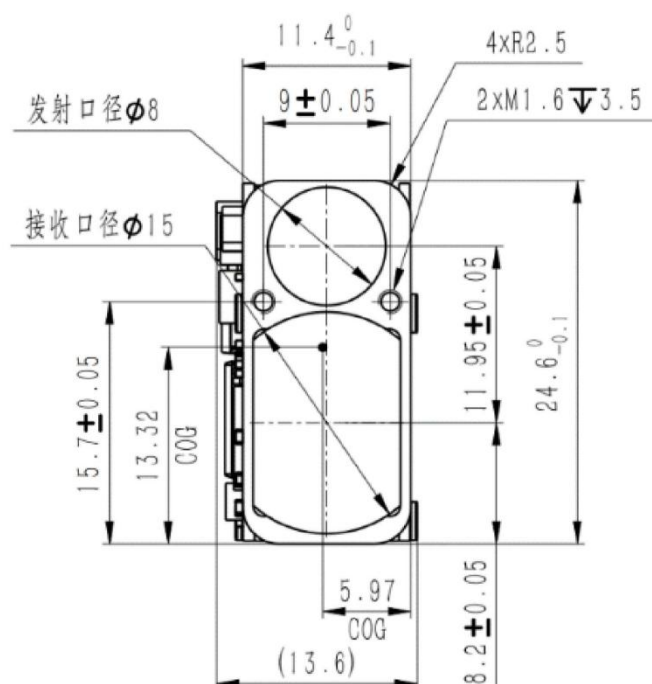
### Cable Definition:



**Table X. Mating Cable Definition:**

| CN1 (C-WB0812H-5-1-S-N) | CN2 (C-WB0812H-6-1-S-N) | Specification | Color  |
|-------------------------|-------------------------|---------------|--------|
| 1                       | 6                       | AWG32         | Black  |
| 2                       | 5                       | AWG32         | Red    |
| 3                       | 3                       | AWG32         | Yellow |
| 4                       | 2                       | AWG32         | Green  |
| 5                       | 1                       | AWG32         | White  |

**Mechanical Interface and Center of Gravity:**



# 05

## Communication Protocol and Commands

**For details of the following contents, please refer to Appendix 1:**

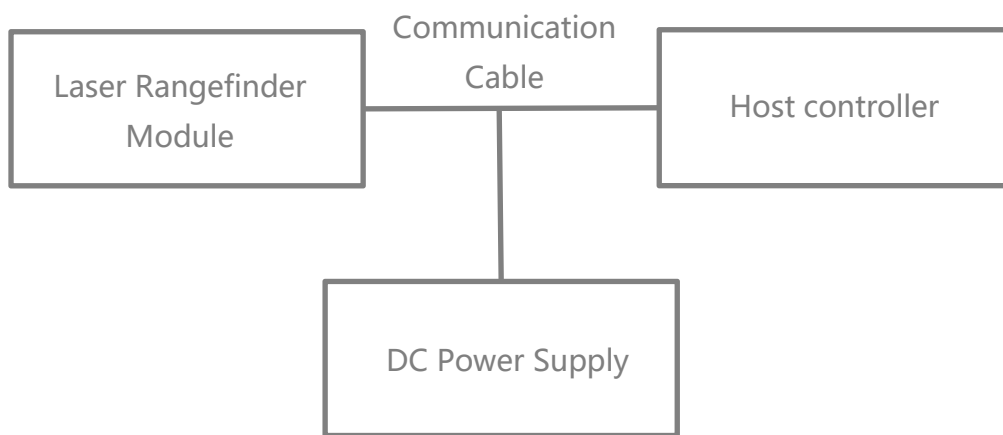
- > Protocol Format
- > Ranging Commands
- > Configuration Commands
- > Other Quick Commands

# 06

## Operating Instructions

### Power-On Procedure

- Before powering on, connect the Laser Rangefinder Module, communication cable, DC power supply, and Host Controller according to the electrical interface requirements and as shown in the figure below.



- The module powers on automatically after the power supply is connected.

### Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

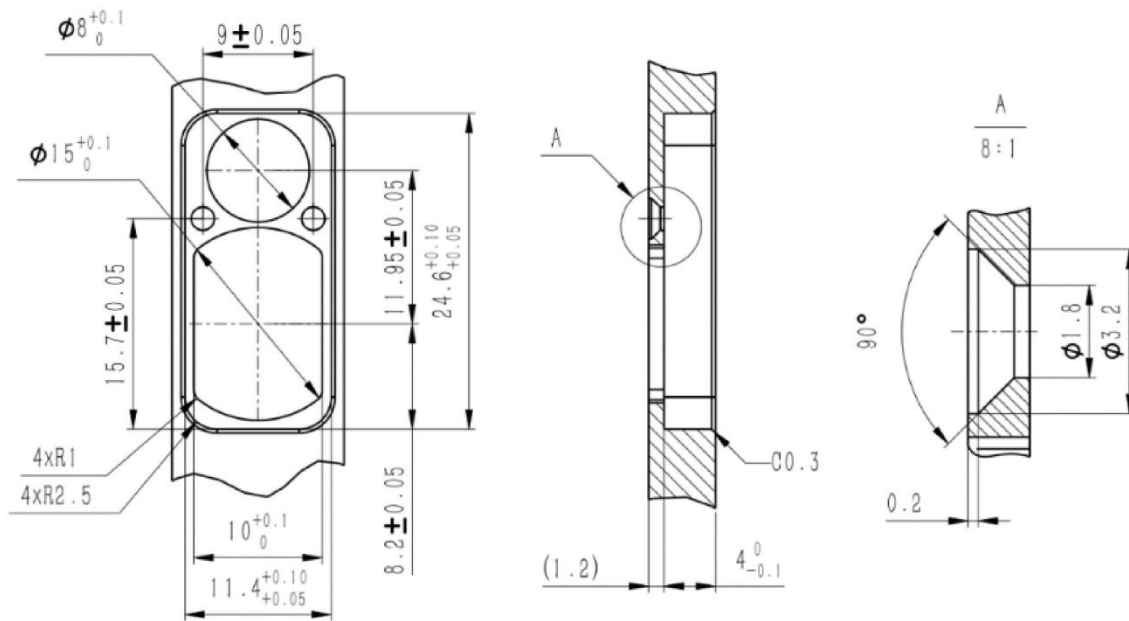
### Ranging Mode Operation

- Send the “Single Ranging” command to the Laser Rangefinder Module. The module performs a single ranging operation and returns the ranging status and distance value.
- Send the “Continuous Ranging” command to the Laser Rangefinder Module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the “Stop Ranging” command to terminate the ranging operation.

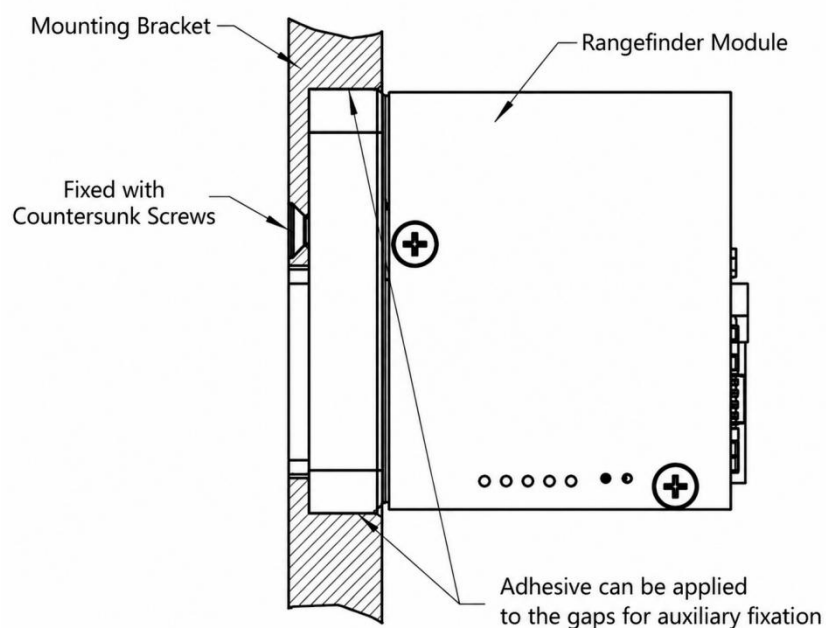
# 07

## Mechanical Dimensions and Mounting Hole Layout

The recommended mounting bracket dimensions are shown below.



### Mounting Diagram:





# 08

## Precautions for Use

- Do not stare directly into the laser emission aperture to avoid eye injury (compliant with Class 1 standards).
- Operating temperature range:  $-20^{\circ}\text{C}$  to  $+70^{\circ}\text{C}$ . Avoid exposure to high temperature, humidity, rain, snow, or dusty environments.
- Do not disassemble the module without authorization, as this may void the warranty or cause device damage.
- This product is mainly intended for integration into civilian UAVs, outdoor observation, engineering surveying, and related equipment. Use in any military applications is prohibited.
- The spot adjustment/debug mode is intended for development personnel only. Do not aim the laser at human eyes while this mode is enabled to avoid eye injury.

# 09

## Optical Window Requirements

- If a protective window lens is used in the final integrated device, it is recommended that the lens transmission at the  $905 \pm 20\text{ nm}$  wavelength band be  $\geq 98\%$ .

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## Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFB2000A Laser Rangefinder Module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

# 11

## Maintenance Instructions

### Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

| No. | Item                      | Quantity | Remarks |
|-----|---------------------------|----------|---------|
| 1   | Laser Rangefinder Module  | 1 unit   | -       |
| 2   | Certificate of Conformity | 1 copy   | -       |
| 3   | Packaging Box             | 1 pc     | -       |
| 4   | Power Cable               | 1 pc     | -       |

## Module Cleaning

### Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

### Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

## Inspection and Maintenance

### Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
  - Check whether the product appearance is normal;

- Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
  - Perform the power-on procedure and verify that the standby current is correct;
  - Start the self-test function of the product;
  - After the inspection is completed, perform the power-off procedure.

### Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
  - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
  - All screws on the panel shall be securely fastened;
  - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance
  - Connect the product power supply in sequence;
  - Perform the power-on procedure and verify that the standby current is correct;
  - Start the product self-test and complete the self-test procedure;
  - Perform the power-off procedure after completion of the inspection.

## Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

| Issue                                                                                           | Possible Cause / Solution                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measured distance is abnormally shorter than the actual distance                                | Testing in heavy haze, fog, or high-humidity environments may cause the module to detect airborne water droplets or particles instead of the actual target. It is recommended to perform testing in clear environmental conditions.                   |
| Ranging performance decreases, or the theoretical target distance cannot be reached             | ① Very low target reflectivity can reduce the returned laser energy and affect ranging performance. When measuring low-reflectivity targets such as lake surfaces, wet roads, or glass, the actual performance may not achieve the theoretical range. |
|                                                                                                 | ② If the ranging module is not aimed perpendicular to the target surface and instead measures at an angle, the returned laser energy will attenuate under oblique measurement conditions, resulting in reduced ranging performance.                   |
|                                                                                                 | ③ Since the laser wavelength overlaps with part of the solar spectrum, direct exposure of the ranging module to sunlight may reduce ranging performance.                                                                                              |
| No response to commands when communicating with a PC using the supplied communication cable set | Check whether the communication cable is damaged or broken (the 6-pin communication cable is relatively thin and can break easily).                                                                                                                   |
| No response to commands when using a custom communication cable                                 | Check whether the external wiring sequence is correct, and verify that the communication cable has no short circuit, open circuit, or poor contact. Also confirm that the UART TX and RX lines are not reversed.                                      |

## Packaging and Transportation Requirements

### > Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

### > Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

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## Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

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## Contact Information

Website: [www.sunflaser.com](http://www.sunflaser.com)

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

## Appendix 1:

### 1. Protocol Format

**Baud Rate:** 115200 bps (factory default) / 57600 bps / 38400 bps / 9600 bps

**Data Format:** 1 start bit, 8 data bits, 1 stop bit, no parity bit

#### 1.1 Data Packet Format

| Field              | Length (Bytes) | Value Range | Description                                                        |
|--------------------|----------------|-------------|--------------------------------------------------------------------|
| Frame Header       | 2              | 0xEE 0x16   | Fixed bytes                                                        |
| Data Length        | 1              | 2–7         | Number of subsequent valid bytes                                   |
| Device Code        | 1              | 0x03        | Fixed byte                                                         |
| Command Code       | 1              | 0x00–0xFF   | Command type identifier                                            |
| Related Parameters | 0–5            | 0x00–0xFF   | Parameter values (depending on the command)                        |
| Checksum           | 1              | 0x00–0xFF   | Sum of Device Code + Command Code + Parameters (lower 8 bits only) |

### 2. Ranging Commands

After the ranging module is powered on, it remains in the default state with the **POWER\_ON** pin pulled low. Before using any of the commands below, the **POWER\_ON** pin must be pulled high and a delay of approximately 0.2 seconds should be allowed for the driver capacitors to fully charge; otherwise, the module may not respond to commands. During subsequent standby operation, if the **POWER\_ON** pin has not been pulled low, the 0.2-second wait is not required.

All commands below are represented in hexadecimal format.

#### 2.1 Device Self-Test

**Send:**

EE 16 02 03 01 04

(Sends the self-test command and returns the current operating status.)

**Response:**

```
EE 16 06 03 01 Status3 Status2 Status1 Status0 Check
```

For bit-level analysis, hexadecimal values should be converted into binary format.

- **Status3:** Reserved
- **Status2:** Reserved
- **Status1:**
  - bit0 — FPGA system status
    - 1: Normal
    - 0: Abnormal
  - bit5 — Bias voltage output status
    - 1: Bias voltage normal
    - 0: Bias voltage abnormal
  - bit6 — Temperature status
    - 1: Temperature normal
    - 0: Temperature abnormal

**Status0:** Reserved

## 2.2 Single Ranging & Continuous Ranging

### ① Single Ranging

**Send:**

```
EE 16 02 03 02 05
```

**Response:**

```
EE 16 06 03 02 Status Result2 Result1 Result0 Check
```

### ② Continuous Ranging

**Send:**

```
EE 16 02 03 04 07
```



**Response:**

```
EE 16 06 03 04 Status Result2 Result1 Result0 Check
```

For bit-level analysis, hexadecimal values should be converted into binary format.

**• Status:**

0x00: Single-target ranging result

0x04: Out-of-range result

- **Result2:** High 8 bits of the ranging value
- **Result1:** Low 8 bits of the ranging value
- **Result0:** Decimal portion of the ranging value

**Distance Calculation Formula:**

$$Distance = Result2 \times 256 + Result1 + Result0 \times 0.1$$

Unit: meters (m)

## 2.3 Stop Ranging

**Send:**

```
EE 16 02 03 05 08
```

**Response:**

```
EE 16 02 03 05 08
```

## 3. Configuration Commands

### 3.1 Set Baud Rate

**Send:**

```
EE 16 06 03 A0 BD24 BD16 BD8 BD0 Check
```

**Response:**

```
EE 16 06 03 A0 BD24 BD16 BD8 BD0 Check
```

For bit-level analysis, hexadecimal values should be converted into binary format.

The baud rate value is of type U32, arranged in big-endian order from **BD24** to **BD0**.

## 4. Other Quick Commands

### 4.1 Enter Spot Brightness Debug Mode

**Send:**

```
EE 16 04 03 10 01 power Check
```

**Response:**

```
EE 16 04 03 10 01 power Check
```

power represents the configured laser brightness percentage, with a valid range of [1–100].

This command affects ranging frequency and eye safety performance. Therefore, it is intended only for spot visualization and debugging purposes. This debug mode must not be enabled during normal ranging operation.

## 4.2 Exit Spot Brightness Debug Mode

(Return to Normal Ranging Mode)

**Send:**

```
EE 16 04 03 10 00 00 13
```

**Response:**

```
EE 16 04 03 10 00 00 13
```

## 4.3 Query FPGA Version

**Send:**

```
EE 16 02 03 A6 A9
```

**Response:**

```
EE 16 06 03 A6 Version1 Version2 Version3 Version4 Check
```

## 4.4 Query MCU Version

**Send:**

```
EE 16 02 03 A7 AA
```

**Response:**

```
EE 16 06 03 A7 Version1 Version2 Version3 Version4 Check
```

## 4.5 Query Hardware Version

**Send:**`EE 16 02 03 A8 AB`**Response:**`EE 16 06 03 A8 Version1 Version2 Version3 Version4 Check`**4.6 Query Device SN****Send:**`EE 16 02 03 A9 AC`**Response:**`EE 16 06 03 A9 SN1 SN2 SN3 SN4 Check`